Super Micro Computer, Inc. (Supermicro) Company

Declaration of Originality

We, the undersigned, declare that the work presented in this project report is the result of our own original efforts and has not been copied, plagiarised, or submitted for academic credit in any other course or institution.

All sources and references used have been properly cited. We understand that any violation of academic integrity may result in disciplinary action by **FEP – Faculdade de Economia da Universidade do Porto**.

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1. **Overview**

Introduction

Super Micro Computer, Inc. (commonly known as Supermicro) is a leading American company specialising in high-performance server solutions, data centre infrastructure, and enterprise computing technologies. Founded in 1993 by Charles Liang and headquartered in San Jose, California, Supermicro has established itself as a key player in the global IT industry. The company focuses on delivering high-efficiency, modular, and scalable server solutions tailored for various industries, including artificial intelligence (AI), cloud computing, enterprise IT, and edge computing.

History and Growth: Since its inception, Supermicro has expanded its footprint worldwide. In 1996, it launched a manufacturing subsidiary, Ablecom, in Taiwan. By 1998, the company had opened a branch in the Netherlands to strengthen its European presence. Over the years, it continued its expansion with a European logistics centre in 2010 and an advanced technology campus in Taiwan in 2012. Today, Supermicro operates on a global scale, providing cutting-edge server technologies to various industries.

Mission and Vision: Supermicro's mission is to provide advanced, high-efficiency computing solutions that reduce energy consumption while maximising computing power. The company envisions a future driven by high-performance computing (HPC), AI-driven applications, and green IT solutions. Sustainability and innovation remain at the core of Supermicro's business strategy, aligning with the increasing demand for energy-efficient and high-performance computing infrastructures.

Industry and Competitive Landscape: Supermicro operates within the highly competitive data centre and enterprise IT sector. The company's primary competitors include:

1. **Dell Technologies** - Known for its PowerEdge server series, Dell is a dominant force in enterprise IT solutions.
2. **Hewlett Packard Enterprise (HPE)** - With its ProLiant server lineup, HPE competes strongly in cloud and AI-based computing.
3. **Lenovo** - Through its ThinkSystem server portfolio, Lenovo has gained a strong market presence.
4. **Cisco Systems** - Provides integrated cloud and networking solutions with its UCS series.
5. **IBM** - Focuses on AI-driven server solutions, particularly in enterprise and cloud applications.
6. **Inspur** - A rising competitor in AI and high-performance computing, particularly in the Chinese market.

Competitive Advantages and Challenges: *Advantages:*

- **Modular Design & Customisation:** Supermicro excels in offering modular and customizable solutions tailored to customer needs.
- **High-Efficiency Computing:** The company is a pioneer in developing energy-efficient server technologies.
- **Innovation in AI & HPC:** Supermicro actively integrates the latest AI and machine learning hardware into its server products.

Challenges:

- **Brand Recognition:** Compared to Dell and HPE, Supermicro has a less established brand name in mainstream enterprise markets.
- **Market Volatility:** The company's reliance on high-tech components makes it susceptible to supply chain disruptions.
- **Regulatory and Compliance Risks:** Supermicro has faced scrutiny over its global supply chain operations, impacting its market reputation.

Recent Stock Market Fluctuations: Supermicro's stock performance has seen significant volatility in recent months. In early 2024, the company's stock surged due to the rising demand for AI-driven server solutions. However, in August 2024, an investigative report by Hindenburg Research alleged that Supermicro engaged in accounting irregularities and potential violations of export regulations related to advanced technology shipments to Russia. This led to a significant stock decline of over 25%.

Further concerns arose when auditing firm Ernst & Young (EY) resigned from its role as Supermicro's financial auditor in October 2024, citing governance and transparency concerns. As a result, the company's market valuation dropped by nearly 30% within a few weeks. While Supermicro has initiated an internal review and reassured stakeholders about its commitment to regulatory compliance, the uncertainty has impacted investor confidence.

Conclusion: Supermicro remains a vital player in the high-performance computing industry. Its focus on sustainability, AI, and energy-efficient computing positions it well for future growth. However, challenges related to regulatory scrutiny and stock volatility could impact its long-term market stability. The company's ability to navigate these obstacles while continuing to innovate will determine its future trajectory in the rapidly evolving tech industry.

2. Financial Analysis

When we look at the Supermicro accounts, we can see that from the period between 2020 and 2024, the company has grown significantly from a financial perspective. This is reflected across the profit and loss account, the balance sheet, and the cash flow statement.
One such indicator of this is the company's net income figures.

Net income

(1000 USD$)	2024	2023	2022	2021	2020
Net income	$1,152,666	$639,998	$285,163	$111,865	$84,308

*(Excel file taken from Orbis)**

Net income rises every year from 2020 and almost doubles in each subsequent year. This rapid growth has been largely attributed to the growth of the AI sector. This net income increase has stemmed from the 110% increase in net sales from 2023 to 2024. The need for increased data centre capabilities has driven up demand for Supermicro's products, which has caused their huge growth in sales.

The impact of the growth in AI is further highlighted when we consider that the COVID period didn't really cause disruption to the company's operations and even saw increased demand for their products due to the move towards a digital economy during this period. This highlights how rapid the growth has been in the AI sector.

Cost of goods sold

Cost of sales and gross margin for fiscal years 2024, 2023 and 2022 are as follows (dollars in millions):

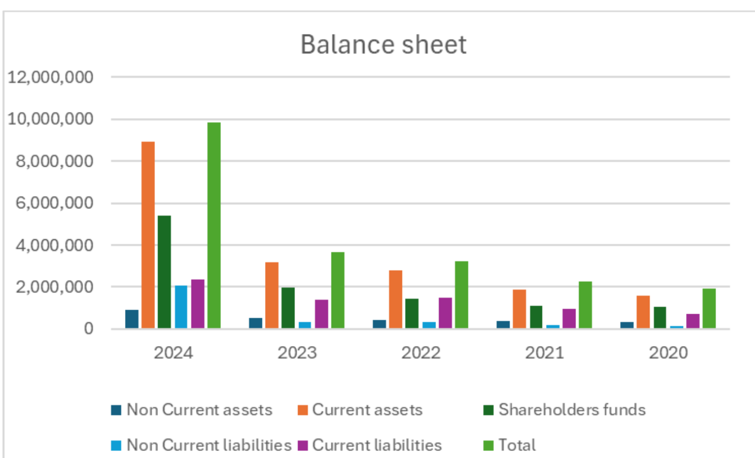
	Years Ended June 30, 2024	2023	2022	2024 over 2023 Change $	%	2023 over 2022 Change $	%
Cost of sales	$ 12,927.8	$ 5,840.5	$ 4,396.1	$ 7,087.3	121.3 %	$ 1,444.4	32.9 %
Gross profit	2,061.4	1,283.0	800.0	778.4	60.7 %	483.0	60.4 %
Gross margin	13.8 %	18.0 %	15.4 %		(4.2)%		2.6 %

*(Cogs, gross profit and gross margin figures from 2022-2024, source: Supermicro annual account 2024)**

The table above shows the increase in cost of goods sold, gross profit, and the decrease in gross margin.

The increased costs from 2023 to 2024 were largely caused by higher component and manufacturing expenses related to increased sales of products that cost more to produce, and there was also an increase in overhead related to operations expansions. These costs, both operational and the cost of goods sold, all reflect the growth of Supermicro and demonstrate the company's healthy financial position. Even the decrease in gross margin was explained as an outcome of a strategic decision by the company to gain market share.

Balance sheet



(*Our own illustration of the balance sheet, figures taken from the Orbis Excel file) **

We can see from the graph above the huge growth from 2023 to 2024 in the balance sheet. Figures on the balance sheet had been growing gradually from 2020 onwards, and then there was a huge jump for 2024. We can see that most of their assets are current assets, which consist mainly of inventories, cash and equivalents, and accounts receivable. For each year up to 2024, non-current liabilities made up a small portion of liabilities, but as we can see in 2024, there is a significant rise in non-current liabilities. This is primarily due to the issuance of convertible notes amounting to $1.6 billion. Current liabilities follow a more stable rise and mainly consist of accounts payable, credit lines, term loans, and accrued liabilities. But as in each of the previous years shareholder funds comprise most of the liabilities of the business, and again there was a significant increase in shareholder funds in 2024, one of the reasons for this rise was a public offering of 20 million shares of common stock, for which they received approximately $1.73 billion, as well as an increase in retained earnings.

Cash Flow

Our key cash flow metrics were as follows (dollars in millions):

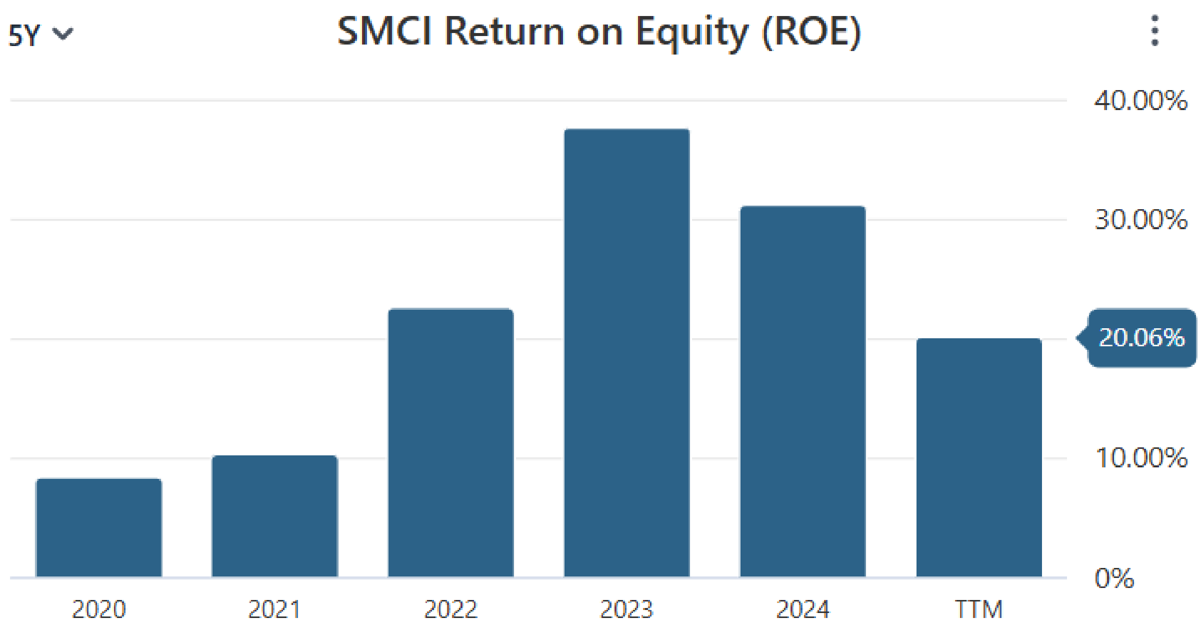
	Years Ended June 30, 2024	2023	2022	2024 over 2023	2023 over 2022
Net cash (used in) provided by operating activities	$ (2,486.0)	$ 663.6	$ (440.8)	$ (3,149.6)	$ 1,104.4
Net cash used in investing activities	$ (194.2)	$ (39.5)	$ (46.3)	$ (154.7)	$ 6.8
Net cash provided by (used in) financing activities	$ 3,911.7	$ (448.3)	$ 522.9	$ 4,360.0	$ (971.2)
Effect of exchange rate fluctuations on cash	$ (2.2)	$ (3.4)	$ (0.7)	$ 1.2	$ (2.7)
Net increase in cash, cash equivalents and restricted cash	$ 1,229.3	$ 172.4	$ 35.1	$ 1,056.9	$ 137.3

*(Cash flow metrics in millions of USD from 2022-2024, source: Supermicro annual report 2024)**

Cash Flow analysis: despite the huge increase in net income, Supermicro's cash flow was -$2.4 billion for 2024, down from $662 billion in 2023. While immediately this seems to demonstrate operational inefficiencies, when we examine the cause, we find that it is attributed to an increase in working capital requirements and a huge increase in inventory. While this may pose a liquidity problem if inventory is not converted into cash or if customer pay times are not properly managed, it doesn't signal an operational issue. The buildup of inventory suggests confidence that the industry will continue to grow, and the sales of the business will keep up with that.

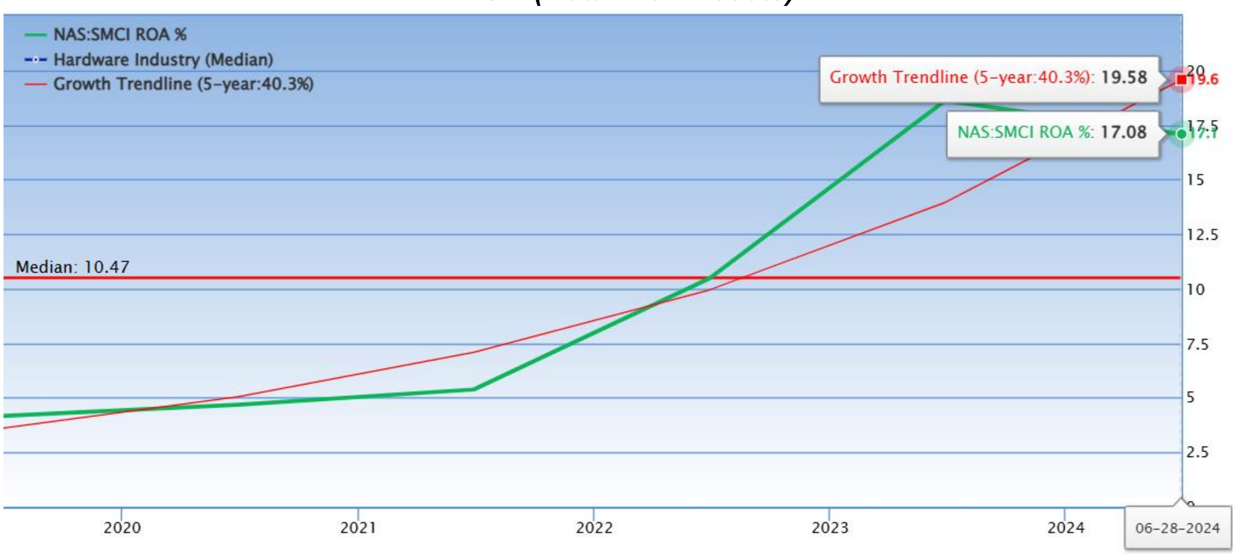
The company's investing activities also demonstrate that they are capitalising on the growth of the sector and are looking to expand in order to keep up with the growing demand. CapEx rose from -$39 million in 2023 to -$194 million in 2024, which marks a huge jump. As this investment activity is leading to negative free cash flow for the business, it is not a sustainable long-term trend, but their operational and investment expenses indicate that they are currently investing to expand operations immediately so that they can manage the growing demand of the sector. Due to the strategic growth planning by Supermicro, their large negative free cash flow will not be an issue as long as they can convert these investments into revenue for the business. This sentiment is backed up by their overall increase in cash and cash equivalents for 2024, where they raised capital through stock offerings and convertible bonds, which signals investor confidence. (Annual Reports, 2025)

ROE (Return on Equity)



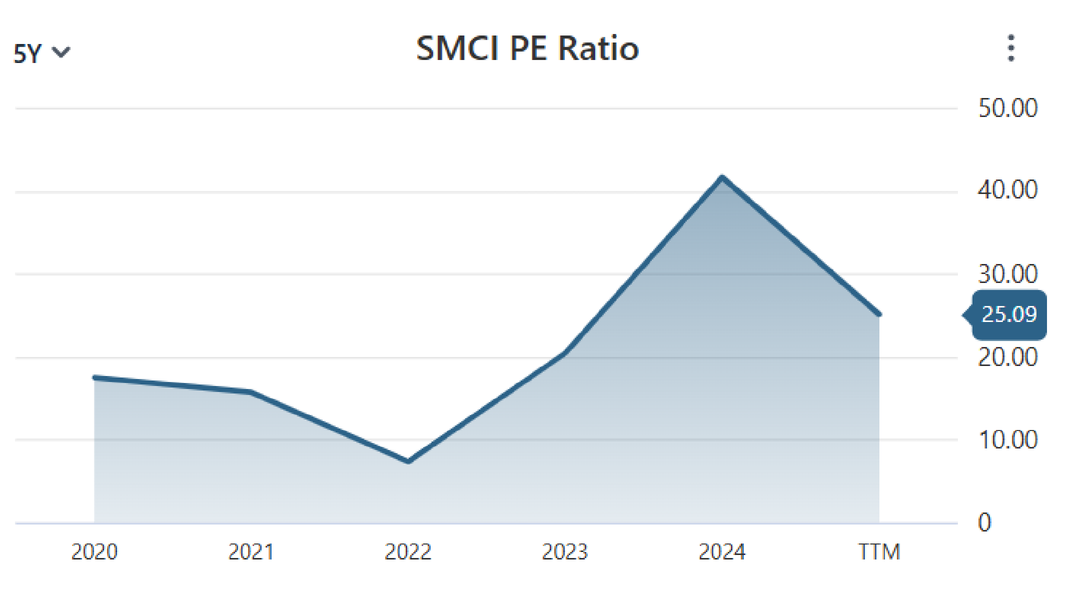
From about 7% in 2020 to a peak of almost 38% in 2023, ROE has demonstrated a robust increasing trend, signifying notable gains in profitability in relation to shareholder equity during this time. The ROE, however, fell to almost 33% in 2024 and then to 20.06% in the past twelve months (TTM), indicating that the company's capacity to produce returns from equity has recently deteriorated, possibly as a result of margin compression.

ROA (Return on Assets)



As of mid-2024, Super Micro Computer Inc. (SMCI) has demonstrated a remarkable upward trend in Return on Assets (ROA), increasing from below 5% in 2019 to 17.08%. This increase shows considerable gains in operational efficiency and is far higher than the hardware sector median of 10.47%. With a compound annual growth rate of almost 40.3% over the last five years, the ROA growth pattern indicates that SMCI has successfully expanded its asset base. This is probably the result of strategic capital investments, rising demand for AI-related infrastructure, and improved cost structure. The latest ROA number, however, is just below the trendline estimate of 19.6%, suggesting that the pace of expansion may be slowing. We use a more cautious long-term ROA assumption of 10% in our DCF model, recognising that the present high efficiency levels might not be permanently viable. Our general judgment that SMCI's present market value may overestimate its long-term cash-generating capacity is supported by this cautious forecast.

PE (Price to Earnings) Ratio



Despite increasing net income ($285M in 2022), the PE ratio fell to its lowest level in 2022, indicating muted investor expectations. But between 2022 and 2024, the PE ratio sharply increased, reaching a peak of over 45x in 2024, which also happened to be the year of a sharp rise in net income to $1.15 billion and robust returns on assets (above 10%) and equity (above 30%). This large increase suggests increased investor optimism, which may be fueled by expansion opportunities in data centres, artificial intelligence, or energy-efficient computers.

It's interesting to note that over the TTM period, the PE ratio fell to 25.09 even though earnings were solid (as evidenced by the 20% ROE and still-high net income), suggesting either:

- Despite impressive earnings, the stock price dropped, or
- In line with dropping ROA and ROE, which indicate a growth of the asset and equity base, earnings expanded more quickly than price.

Given that ROE and ROA have both fallen from their 2023 heights, this decrease in PE despite strong earnings may be the result of market reevaluation of future growth potential or sustainability concerns. In summary, following a phase of rapid expansion and investor fervour, SMCI's valuation seems to be returning to normal.

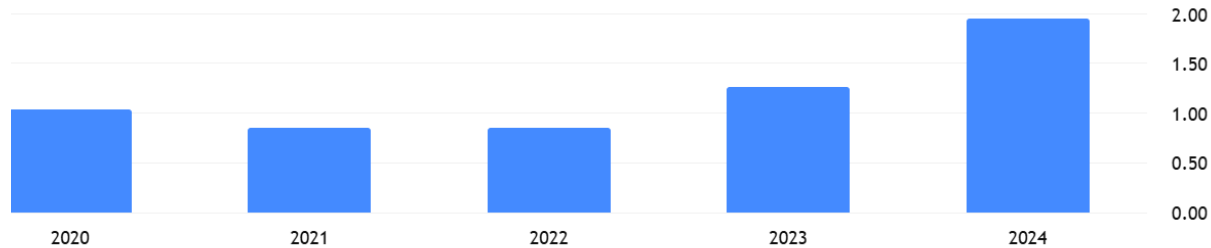
Gross margin

	2024	2023	2022	2021	2020
Gross margin (%)	14.03	18.50	16.02	15.82	16.61

*(Excel file taken from Orbis)**

Another financial aspect that the company highlights is its gross margin, which tells us how much of each sales dollar the company keeps after accounting for operating expenses. As we can see, the figure dropped from 2023 to 2024, but this was intentional as it was the result of Supermicro's strategy to gain a greater market share by offering competitive pricing. They also changed their product and customer mix, focusing more on large enterprises and large data centre customers. They also faced increased costs related to manufacturing.

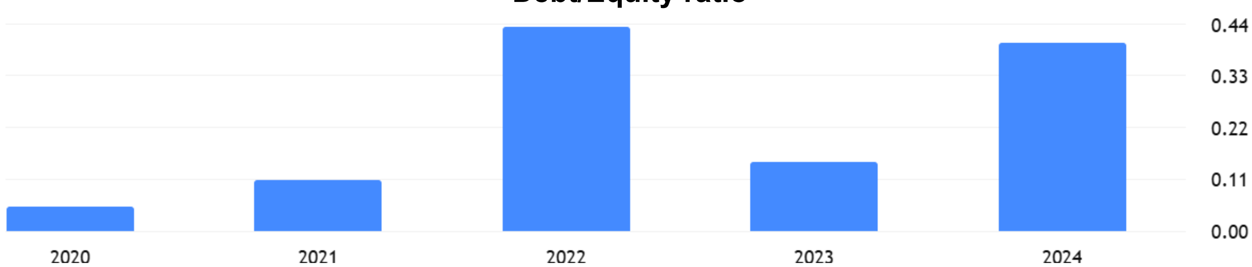
Quick ratio



(Figure taken from Tradingview)

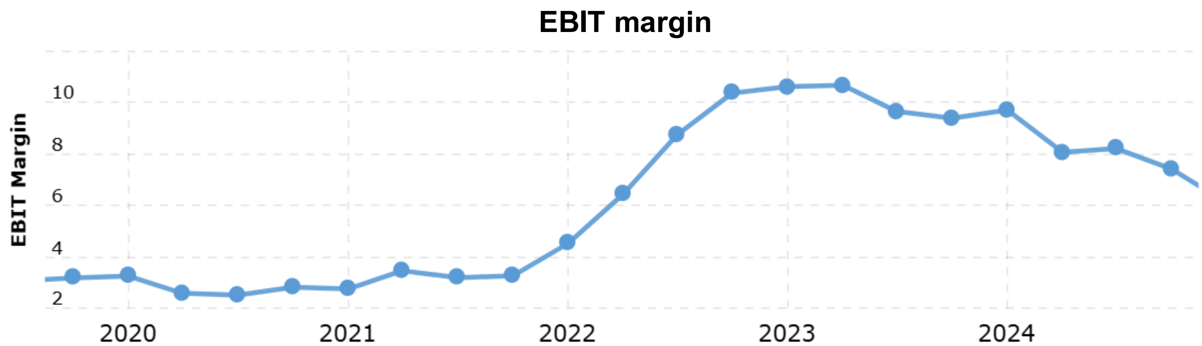
Quick ratio gives us a better indication of Supermicro's liquidity position than the current ratio, as it excludes its huge inventory acquisition for 2024, and inventory is not always very liquid in the hardware manufacturing industry. It rose from 1.26 to 1.96 in 2024 despite an increase in current liabilities from $1.3 billion to $2.3 billion. This is largely due to the huge increase in accounts receivable over that period, which is not as liquid as cash. While this means that currently they have a healthy liquidity position in terms of current assets to match current liabilities, they will have to tightly monitor their collection period, which has been growing over this time and is currently 66 days.

Debt/Equity ratio



(Figure taken from Tradingview)

Supermicro's debt/equity ratio is quite low at 0.4 for 2024, which tells us that they rely heavily on equity financing and don't use much debt. When compared to the debt/equity ratio for other companies operating in the technology and hardware sectors, we see that they fall in line with their peers. (Sarath, 2025). Although we can see the level of debt has risen since 2020, in line with the company's growth, it seems that they are more content to raise capital through equity, and this perhaps could be due to the rapidly changing technology industry, which can be volatile, and so equity seems a safer choice of financing.



(Figure taken from Macrotrends)

EBIT margin is a profitability ratio that tells us what percentage of revenue is left after all operating costs are accounted for. Supermicros EBIT margin has grown from 2020 to 2024 due to the increase in sales from the growth of the AI industry but it fell from a peak of 10.68% in 2023 to 7.42% in 2024, this means that for 2024 less revenue was kept after considering operating expenses, this is likely due to their current investment efforts to expand operations which while currently damaging their EBIT margin could improve it in the future.

Initial diagnosis:

- **Healthy Financial position:** From the brief financial analysis we conducted, we believe that the company displays a healthy financial performance in almost all of the KPIs that we have looked at so far. Although we will need to conduct a more expansive analysis beyond our intermediate report for now, we believe that the company is showing signs of sustained growth, which will continue to positively affect the company's financial records.

- **Importance of AI:** As mentioned multiple times in Supermicro's own financial reports, their growth over the past number of years is mainly attributed to the growing Artificial Intelligence sector. We believe that Supermicro has positioned itself to be one of the biggest players in supplying the hardware necessary for the continued growth of this industry, which will go hand in hand with the company's growth.

- **Market volatility:** as we mentioned the company has faced volatility in its stock price due to concerns over their regulatory compliance and alleged accounting irregularities, these drops in stock price coupled with the company's growth has made it a volatile stock in the market and this is an aspect the company needs to look at to sooth investors' concerns so that the company can see continued investment and growth on the stock market.

3. Valuation

To get an estimate of firm value, we will use two different methods. Firstly, relative valuation to compare Supermicro to similar companies within their industry, relative valuation will use market multiples to determine how the market values Supermicro. This method, however, has its downsides as it assumes markets efficiently and accurately determine company value, which is not always the case, and secondly, it is based on estimates and forward-looking scenarios, which can be hard to determine accurately for a fast-growing company like Supermicro operating in a rapidly growing industry such as AI. This is why we will also be using a discounted cash flow model, which determines company value based on its fundamentals and potential future cash flows. This method also has its limitations, as it requires many assumptions to complete the valuation.

Relative valuation: To complete a relative valuation, we have to determine a peer group of companies operating in the same industry. The companies we have chosen to compare market multiples with are HP Enterprises, Dell, Lenovo, and Cisco. These are some of the most relevant competitors for Supermicro in the data centre hardware and enterprise IT infrastructure, despite differences between them. Dell is a much larger company than Supermicro, but they both are competing in the same core market with overlapping products, meaning they often compete for the same customers. Dell has a larger range of products and is a more established brand, which explains its scale compared to Supermicro, but like Supermicro, they have been benefiting from the drive in demand for AI (Davis, 2025). HPE is also a huge player in the server industry and has seen even further growth in the past few years due to the growth of AI (Westfall & Newman, 2025). Lenovo and Cisco are also comparable to Supermicro due to their similar product range and focus on AI infrastructure. As we identified in the initial diagnosis, the expansion of AI is the most relevant aspect for Supermicro right now, and each of these companies is also heavily influenced by this factor, which is why we believe they make up a good peer group for relative comparison.

Companies	share price($) close of day 16th May 2025	# shares(m)	debt($m)	equity($m)	EBIT($M)	EBITDA($m)	net income($m)
Supermicro	**46.15**	**596.82**	**2650**	**5417**	**1328**	**1380**	**1150**
HP	17.75	1310	17920	24880	2420	4926	2760
DELL	114.19	344.35	25370	-1387	6462	9585	4590
Lenovo	25.27	620.23	3980	6081	2338	3766	1540
Cisco	63.62	3980	29280	45457	11380	14616	9790

(Figures taken from Yahoo Finance, except equity, which is taken from each company's annual reports)

The first table highlights some key financial metrics of the peer group we have selected, which will be relevant for our calculations.

Then, in Table 2, we present the ratios we found for each of the companies that we used to compute an average and median value for the sector that we applied to Supermicro's financials in order to determine their share value according to these average and median values.

Company	Forward PE ratio	EV/EBITDA	PBV	PEG ratio
Supermicro	**13.57**	**19.93**	**4.32**	**0.76**
HP	9.44	5.64	1.12	4.55
DELL	12.25	10.5	-53.4	0.88
Lenovo	9.15	4.17	2.85	0.56
Cisco	16	17.7	5.54	2.13
Avergae	**11.71**	**9.50**	**-10.97**	**2.03**
Median	**10.85**	**8.07**	**1.99**	**1.51**

(Figures taken from Yahoo Finance, except Supermicro PEG, taken from Wealthyhood)

SMCI share value according to:	**PE**	**EV/EBITDA**	**PBV**
Average	$22.56	$17.53	-$99.59
Median	$20.90	$14.22	$18.02

(Share price according to market multiples)

For the PBV ratio, Dell's huge negative PBV value has skewed the result for both the median and average. We have left it in as the median value lines up with our other ratios for pricing Supermicro. At first glance, all the ratios appear to tell a similar story, which is that Supermicro is largely overvalued by the market. When we look at the share price according to median P/E, we see Supermicro is at $20.90 when compared to its actual price of $46.15 this means Supermicro is overvalued by double its actual worth. However, when we look at Supermicro's PEG of 0.76, which is lower than the median of 1.5, we get a different story. The PEG figure tells us that Supermicro is cheap relative to its growth potential and that the market's valuation is based on expected growth in earnings, which aligns with its current growth trajectory based on the growing demand for AI infrastructure.

However, the flip side is that Supermicro could just be an overvalued share if they don't meet the expected growth. Also, given the previously mentioned volatility of Supermicro's share price in recent times, perhaps the current share price is just a reflection of that volatility.

EV/EBITDA gives the same indication, which is that Supermicro is heavily overpriced with an actual value of $14.22. This tells us that, based on operating income, Supermicro is not worth the market price, but the issue is that this valuation does not capture its growth potential. Supermicro's EBITDA is suppressed relative to its market cap, which is very high due to growth expectations. When we take their EV/EBITDA value on its own, we see that at 19.93, it is much higher than the median of 8.07, indicating that investors are willing to pay more for each dollar of EBITDA, possibly due to growth expectations.

And finally, for the share price according to the median PBV, again we see an indication that Supermicro is overvalued at $18.01. Again, Supermicro's PBV at 4.32 is above the median of 1.99, which tells us that the market values its equity highly, probably due to its growth potential. Overall our relative valuation based on the chosen market multiples signals that Supermicro is highly overvalued, however the problem with these multiples is that except for PEG they don't account for growth potential of the company and as we identified in our initial diagnosis Supermicro is riding the wave of surging AI demand which could potentially generate huge revenue growth for the company into the foreseeable future as it has done over the past few years if things continue to go well. Our analysis suggests that Supermicro is an outlier itself in the hardware and server industries due to its high growth potential, and we believe this is why the market is pricing it so high above what our market multiple values suggest.

In order to get another valuation of Supermicro, we did a discounted cash flow analysis to get a valuation based on the intrinsic value of the firm.

Model Choice:
To determine the intrinsic value of Super Micro Computer Inc. (SMCI), we chose to apply a Discounted Cash Flow (DCF) model, as it provides a company-specific, forward-looking approach based on expected free cash flows. This strategy is appropriate for evaluating SMCI's long-term value development because it is a growth-oriented company in the high-tech server and infrastructure sector.

Several hypotheses are needed for the model, especially regarding tax rates, reinvestment, predicted earnings growth, and the cost of capital. We made the following assumptions based on SMCI's past performance, the macroeconomic environment, and analyst expectations:

- 1.49% average EBIT growth during the projection period
- 8.71% reinvestment rate based on FCFF structure and SMCI's CAPEX trends
- 17.08% return on capital (ROIC), which represents the efficiency of existing operations
- The steady innovation and scalability of SMCI justify a terminal growth rate of 2.5 per cent, which is marginally higher than long-term US inflation.

Capital Costs:
It is crucial to assess the company's capital structure and ascertain the costs of debt and equity to compute Super Micro Computer Inc.'s weighted average cost of capital, or WACC.

According to Damodaran's industry data, we take the unlevered beta for the server and hardware sector to be 1.00. Based on SMCI's reported income tax expense in relation to pre-tax income, the effective tax rate for 2024 is projected to be 5.21%.

Furthermore, we assume that the market capitalisation (equity value) is $48.18 billion, based on a stock price of $81.23 and 588.09 million outstanding shares, and that the market value of debt is around $2.17 billion, which is the same as the book value of debt.

Using the Capital Asset Pricing Model (CAPM):

Cost of Equity

Using the Capital Asset Pricing Model (CAPM):

$ke = rf + \beta L \cdot MRP$

$ke = 4.46\% + 1.36 \cdot 5.50\% = \%11.94$

Cost of Debt

To estimate the cost of debt, we analyse the company's **interest coverage ratio**:

Interest Coverage Ratio = EBIT /Interest Expense

Interest Coverage Ratio = 1,233,490,000 / 5,933,000≈208

According to the NYU Stern default spread table, an interest coverage ratio above 8.5 corresponds to a **rating of Aaa/AAA** with a **default spread of 0.45%**.

For large non-financial service firms

If interest coverage ratio is			
>	≤ to	*Rating is*	*Spread is*
-100000	0.199999	D2/D	19.00%
0.2	0.649999	C2/C	15.50%
0.65	0.799999	Ca2/CC	10.10%
0.8	1.249999	Caa/CCC	7.28%
1.25	1.499999	B3/B-	4.42%
1.5	1.749999	B2/B	3.00%
1.75	1.999999	B1/B+	2.61%
2	2.2499999	Ba2/BB	1.83%
2.25	2.49999	Ba1/BB+	1.55%
2.5	2.999999	Baa2/BBB	1.20%
3	4.249999	A3/A-	0.95%
4.25	5.499999	A2/A	0.85%
5.5	6.499999	A1/A+	0.77%
6.5	8.499999	Aa2/AA	0.60%
8.50	100000	Aaa/AAA	0.45%

Table 8: Default Spread Table (Data derived from NYU Stern)

$Kd = rf + \text{Default Spread}$

$Kd = 4.46\% + 0.45\% = 4.91\%$

After-tax cost of debt:

$kd(1-T) = 4.91\% \cdot (1-0.0521) \approx 4.65$

WACC Calculation

WACC is calculated as follows:

$WACC = (E / (D+E)) \cdot ke + (D/(D+E)) \cdot kd(1-T)$

WACC=(48.18B/(2.17B+48.18B))·11.94%+(2.17B/(48.18B+2.17B))·4.65%

WACC≈11.63%

The terminal WACC is calculated with a levered beta of 1 and a higher effective tax rate (21%):

WACC terminal ≈9.70%

Discounted Cash Flow Valuation

	2024	2025	2026	2027	2028	2029	Terminal
EBIT	$1,233,490,000	$1,263,280,165	$1,293,789,796	$1,325,036,268	$1,357,037,378	$1,389,811,351	$1,544,836,175
EBIT(1-T)	$1,169,225,171	$1,197,463,268	$1,226,383,348	$1,256,001,879	$1,286,335,731	$1,317,402,180	$1,220,420,578
Reinvestment		$169,321,306	$173,410,605	$177,598,666	$181,887,872	$186,280,668	$61,021,029
FCFF		**$1,028,141,962**	**$1,052,972,742**	**$1,078,403,213**	**$1,104,447,859**	**$1,131,121,511**	**$1,159,399,549**
Cot of capital		11.63%	11.63%	11.63%	11.63%	11.63%	9.70%
Discount factor		0.8959	0.8026	0.7190	0.6441	0.57701	
PV (FCFF)		**$921,064,254**	**$845,066,363**	**$775,339,130**	**$711,365,158**	**$652,669,739**	

(DCF model taken from Excel file)

By projecting unlevered free cash flows (FCFF) and discounting them at the weighted average cost of capital (WACC), the DCF model offers an organised evaluation of the value of a company. In order to account for any future cash flows beyond the explicit prediction, we first projected SMCI's financials over a 5-year horizon (2025–2029), and then we included a terminal period.

Predicting FCFF

To calculate FCFF, we started with EBIT, deducted taxes using the declared effective tax rate (5.21%), and then subtracted reinvestment.

Reinvestment Rate = $(CAPEX - Depreciation)$ + ΔWC / EBIT * (1 - Tax Rate)

Because of the enormous amount of negative change in Working Capital, it is assumed to be zero.

Reinvestment Rate = (-124.28M - 40.99M)/ 1233.49 * (1 - 0.0521) = 14.14%

g= Reinvestment Rate x ROA

g= 17.08% x 14.14% = 2.42

This derived growth rate closely aligns with the **terminal growth rate assumption of 2.5%**, reinforcing the model's logic and validating our long-term expectations for SMCI's cash generation under its current investment intensity.

	2024	2025	2026	2027	2028	2029
EBIT	$1,233,490,000	$1,263,280,165	$1,293,789,796	$1,325,036,268	$1,357,037,378	$1,389,811,351
EBIT(1-T)	$1,169,225,171	$1,197,463,268	$1,226,383,348	$1,256,001,879	$1,286,335,731	$1,317,402,180
Reinvestment		$169,321,306	$173,410,605	$177,598,666	$181,887,872	$186,280,668
FCFF		**$1,028,141,962**	**$1,052,972,742**	**$1,078,403,213**	**$1,104,447,859**	**$1,131,121,511**

FCFF analysis from our Excel File

The terminal value was calculated using a perpetual growth model:

Terminal Value= FCFF 2029 x (1 +g) /(WACC terminal - g)

Terminal Value = 1,131,121,511 x 1.025 / (0.097 - 0.025)

Terminal Value = $16,108,483,725

FCFF (terminal)	$1,159,399,549
Cost of capital (termi	9.70%
Value (terminal)	$16,108,483,725
PV (Terminal value)	$10,140,796,432
PV (initial FCFF)	$3,905,504,644
EV	$14,046,301,076
Minority holdings	$0
Cash	$1,670,273,000
Firm value	$15,716,574,076
- Minorit interest	$164,000
- Debt	$2,174,150,000
Equity	$13,542,260,076
Nº shares (m)	588,090,000
Valur per share	**$23.03**
Price	$81.93
Price/Value	355.79%

(DCF model from Excel file)

Based on the terminal value calculated from the DCF, we found a share price of $23.03, which is far below the market price of $81.93 from the time the figures used to calculate the DCF were taken. Even now, with a share price of $46.15(close of day May 16th), the market still values it at double the price.

4. **Forward-looking scenarios**

Although things seem to be heading in a positive direction for Supermicro with massive growth over the past few years and the potential for even further growth, which seems to be the current market sentiment, given the results of our relative valuation. As we mentioned, this growth is largely driven by the increased demand for AI infrastructure. Supermicro's position as a hardware and server supplier for this industry means that as the growth of this industry continues, so will the company's own growth.

However, there is potential for things to go wrong. Factors such as the recent allegations of irregularities in their accounts and the departure of EY as their financial auditor due to governance and transparency concerns could lead to problems for the business. Supermicro could be hit with punishments such as fines and further scrutiny of their practices, which could cause investors to lose confidence and a decline in their stock price, which has already faced huge volatility related to these issues.

Another potential scenario is the decrease in AI spending. Supermicro's recent growth has been a result of the AI surge, and its massive investment in growth is a gamble that this trend will continue and that AI is the future of the company, but if a situation arises in which AI is replaced with other technologies, the company could be in trouble. This potential situation aligns better with our relative valuation calculations of Supermicro's share price and perhaps demonstrates that they are a significant outlier in the industry, as more than any other company, they are dependent on this growth to continue in order to remain competitive.

Sensitivity analysis

		Terminal WACC				
	$23.03	6%	7%	9.70%	9%	10%
	1.50%	$38.20	$31.09	$20.78	$22.70	$20.04
	2.00%	$42.43	$33.76	$21.83	$24.00	$21.01
	2.50%	$47.88	$37.02	$23.03	$25.50	$22.11
	3.00%	$55.13	$41.10	$24.41	$27.24	$23.36
g	3.50%	$65.29	$46.34	$26.01	$29.31	$24.80

(taken from Excel file)

Sensitivity analysis: We conducted a sensitivity analysis based on our DCF model to get a better understanding of how different scenarios would affect the share value. To do this, we calculated the share price based on different possible values for the weighted average cost of capital, which was the discount factor used to calculate the FCFF and different levels of growth. We calculated the worst-case scenario of a 1.5% growth rate with a discount factor of 10%, which gets us a share price of $20.04. An optimal scenario of 2.5% growth rate with a weighted average cost of capital of 8.22%, which gets a price of $23.03, and the best-case scenario, where growth is at 3.5% and a discount of 6%, where we get a share price of $65.29. Even in the best-case scenario, Supermicro's share price is far below its actual trading price of $81.23 at 1st of July 2024, where we got our figures to calculate the DCF from. Even now, with nearly a year to grow, although Supermicro's price has dropped over this time, it is still above our optimal share price scenario.

5. Conclusion

Both our relative valuation and the discounted cash flow indicate that Supermicro is overvalued. If we take the relative valuation in isolation, we could argue that, due to the growth potential of the company, its market value is fair, as investors are willing to pay for potential and relative valuation only provides a snapshot of the company at a moment in time. This is why the discounted cash flow is needed to get a better understanding of a growth company like Supermicro. As our model shows, though, even when we take into account growth for the company and consider its own characteristics and intrinsic value, we still find that it is significantly overpriced.
The results of the DCF also justify the relative valuation results, as they are much closer in value to each other than the actual price. Despite only a small difference between the P/E share price of $20.90 and the DCF price of $23.03, we believe that the DCF is a better model for Supermicro as it is better at accounting for the company's growth.
While our model anticipates that FCFF will continue to grow in the future, it does not seem to be at the rate of growth that investors appear to be expecting.

Overall, given the sharp discrepancy between the market price and our calculated intrinsic value, we do not recommend investing in SMCI at its current valuation. Supermicro has strong growth potential, which is why investors can still invest in Supermicro if they are willing to take risky investments for growth opportunities, the margin of safety is absent, and the risk of correction outweighs the opportunity for upside. Until cash flow growth and returns justify its premium pricing, SMCI appears to be overvalued, and investors should remain cautious.

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